back and forth and the Z axis moves up and down. While the X and Z axis movements are executed using the same structure, the Y axis, which also doubles up as the base moves independently. All three axes are mounted on rails and run over them so the axes need to be mounted on brushes.

The brushes can either be printed or the sturdier alternative would be to use brass brushings.

Interesting fact: Brass has the least friction and is self lubricating. Whodda thunk it?

The X, Y and Z axes are controlled by stepper motors, with the Z axis requiring two of them. While it is up to the builder to choose the specifications of the motors for each axis, we and the rest of the RepRap community would advice that you stick to two motors of the same specs when it comes to the Z axis.

In the case of the Prusa Mendel, the X,Y and Z axes can travel across the work space and print an object up to 200mm x 200mm x 100mm in length, breadth and height.

The Extruder

'To extrude' means to thrust or force out, so this part's function is sort of self explanatory. The extruder is the money maker when it comes to a 3D printer. It is made up of a brass bolt with a hole drilled through it. The hole is around 3mm at the end through which the filament is fed through and the melted filament comes out the other end through a hole that is less than a millimetre across.

The Business End a.k.a The Extruder

The Prusa Mendel's extruder is called the Wade's extruder or a Wade's geared extruder. It comprises of a hot part and a cold part. The cold part is separated from the heat by a heat shield so that the filament doesn't melt when it doesn't need to. While it's possible to build an extruder from scratch, it's a lot easier to pick one up online. Ebay has extruders for the Prusa Mendel in a price bracket of $25 to about $70 if it comes with a motor for controlling the extrusion. $13 if you want to piece one together though.

An interesting thing that comes to mind here is the 3Doodler, a 3D print pen which extrudes plastic and allows the user to 'doodle' in 3 dimensions.